# A Sparse-Sensing Spatial Digital Twin for Learning Environmental Impacts of Non-Networked Appliances in a Single Room

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Abstract—Practical room-scale digital twins must often operate with sparse sensors and non-networked appliances that expose no telemetry. This paper presents a single-room spatial digital twin that estimates temperature, humidity, and illuminance fields from eight corner sensors while modeling conventional air conditioners, manual windows, and ordinary lights. The proposed design combines an appliance-aware bulk-plus-local field model, sparse calibration, appliance-impact learning, and an optional hybrid residual neural layer. Across eight leave-one-scenario-out folds, average field MAE is 0.1723/0.4633/54.9052 for IDW, 0.0474/0.1765/2.0269 for the base estimator, 0.2091/0.2241/48.1422 for a standalone pure RNN without physics estimates, and 0.0017/0.0059/0.1407 for the hybrid; the pure RNN is lowest in 0 of 24 fold-factor comparisons. On a preliminary seven-day bedroom dataset, calibration reduces held-out pillow-point MAE from 0.8967/4.1286/309.0142 to 0.1676/0.3939/16.6450, with positive date-block bootstrap intervals and date-deletion reductions. The evaluation separates controlled full-field evidence, public task-aligned comparisons, and real sparse-calibration checks, supporting complementary roles for physical structure, calibration, and residual learning.

Index Terms—spatial digital twin, sparse sensing, indoor environment modeling, non-networked appliances, reduced-order model, sensor calibration

## I. Introduction

Smart homes and smart buildings require indoor environmental awareness for comfort assessment, energy management, and device control. A practical system should estimate conditions at relevant regions such as a room center, window-side zone, or bed location, not only at installed sensors. This is difficult because dense sensing is intrusive, while ordinary rooms also contain conventional appliances that do not report state, power, or operating parameters.

This work targets a common but under-instrumented digital-twin setting: the room contains influential physical assets, but those assets are not necessarily connected to a building management system. The twin must therefore infer appliance state effects from sparse environmental observations, support what-if queries, and keep its claim boundary clear. Pure interpolation ignores appliance geometry and directionality, while a purely local field model misses whole-room settling after air-conditioner activation. We therefore use a reduced-order appliance-aware model as the primary estimator and evaluate, rather than deploy, an end-to-end pure RNN field baseline alongside structured residual correction.

The contributions are: (i) a three-factor room digital twin with bulk-plus-local appliance influence functions, photometric lamp falloff, and one-bounce diffuse illuminance reflection; (ii) a sparse-sensor calibration pipeline combining active-device power fitting, trilinear residual correction, and before/after impact learning; (iii) a hybrid residual pathway that preserves the reduced-order estimator while refining systematic error; and (iv) an implementation and validation hierarchy covering synthetic full-field scenarios, ablations, leave-one-scenario-out residual learning, window-condition cases, public task-aligned datasets, and preliminary real-bedroom snapshots. Figure 1 summarizes the argument chain from the research gap through methods and evidence to the claim boundary; it also keeps the MCP-compatible interface secondary to the modeling contribution and marks intervention-based recommendation validation as future work.

## II. Related Work

Digital twins increasingly represent physical systems in manufacturing, infrastructure, and smart buildings [1], [2]. Building reviews emphasize BIM/BMS connectivity, while sparse-sensor spatial estimation remains less explored [3]. Reduced-order and grey-box models are interpretable alternatives to CFD [4], [5]; zonal and hybrid models

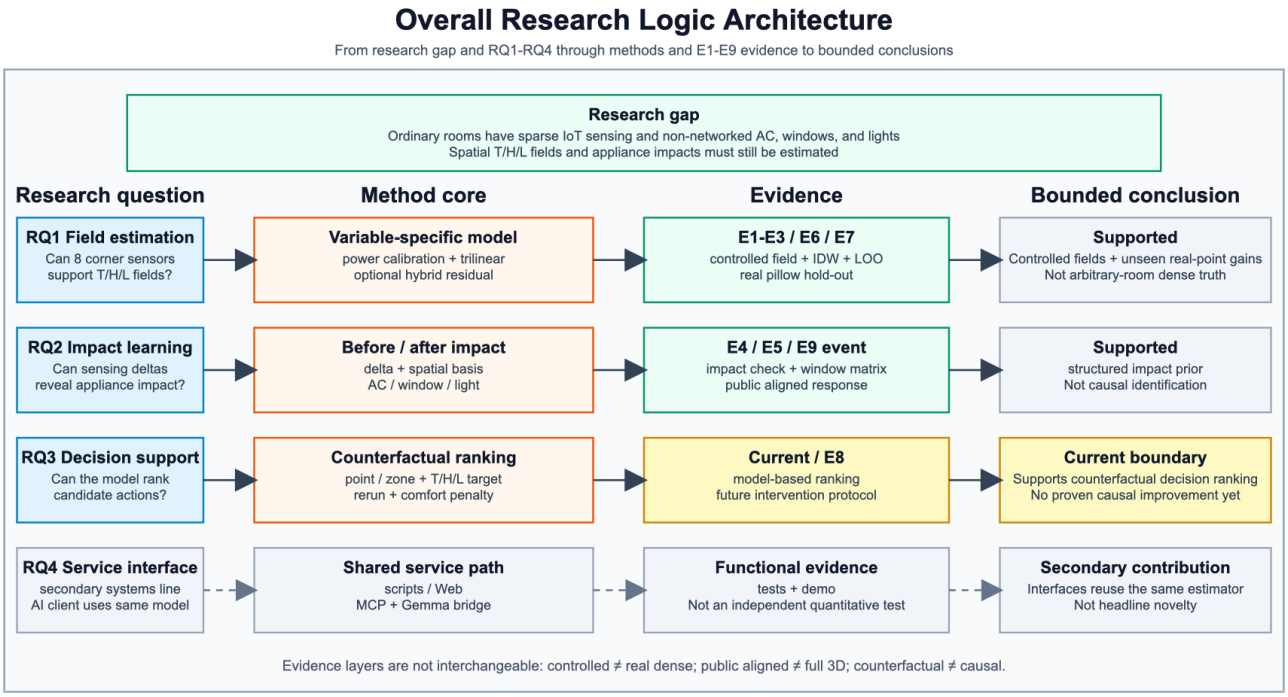



Fig. 1. Overall argument map linking the sparse-sensing and non-networked-appliance gap to RQ1–RQ4, method components, the E1–E9 evidence hierarchy, and bounded conclusions. RQ4 is a secondary service line, while E8 remains a future intervention protocol.

retain dominant non-uniformity at lower cost [6], [7]. Public BMC telemetry exposes temperatures, fan speeds, and power [8]. Hybrid forecasting can learn simulation–measurement residuals [9], and data assimilation shows that finite observations and sensor placement matter [10], [11]. This work couples sparse sensing with appliance-aware influence bases and residual calibration; IDW [12] is the geometry-free baseline. Local APIs, a web demo, and an MCP-compatible layer expose the estimator but remain secondary [13].

## III. Method

### A. Room, Devices, and Field Model

The canonical room is a 6.0 m by 4.0 m by 3.0 m rectangular room with eight corner sensing nodes, four on the floor and four on the ceiling. Each node measures temperature, humidity, and illuminance. The room contains three evaluation zones and three appliance classes: an air conditioner, a window, and a light. The web demo represents these appliances as wall or ceiling markers and allows their activations to be toggled.

For metric $m \in \{T, H, L\}$ at point $\mathbf{p} = (x, y, z)$ and elapsed time $t$, the estimated field is

$$F_m(\mathbf{p}, t) = B_m^{\text{bulk}}(t) + B_m^{\text{local}}(\mathbf{p}, t) + \sum_j I_{j,m}(\mathbf{p}, t) + C_m(\mathbf{p}). \tag{1}$$

The bulk term captures whole-room settling, local terms preserve spatial gradients, $I_{j,m}$ represents appliance influence, and $C_m$ is fitted from sparse sensor residuals. Each appliance influence is modulated by

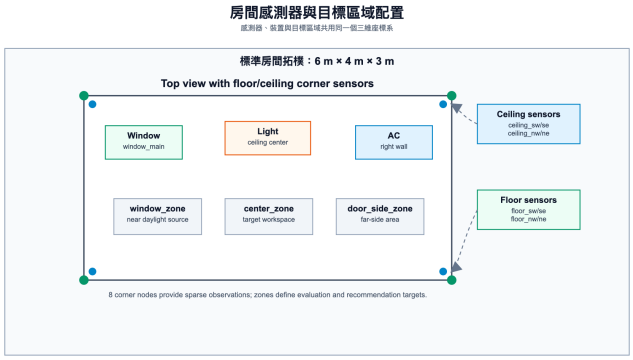



Fig. 2. Room topology, appliance placement, target zones, and the eight floor/ceiling corner sensors.

$$E_j(\mathbf{p}, t) = a_j(1 - e^{-t/\tau_j})D_j(\mathbf{p})G_j(\mathbf{p}), \tag{2}$$

where $a_j$ is activation, $\tau_j$ is a response time constant, $D_j$ is distance attenuation, and $G_j$ is directional gain. The air conditioner lowers temperature and humidity, the window uses outdoor temperature, humidity, and sunlight, and the light primarily raises illuminance with a small heat gain. The lamp direct term uses a cosine beam weight derived from beam angle, an inverse-square distance scale with bounded gain, and line-of-sight obstruction. Window daylight remains an envelope-based daylight term in the

reported experiments; an aperture view-factor option is implemented but not used as the default because it increased window-family illuminance error in the current canonical setup.

Direct window and lamp terms underestimate indirect fill light near floors, walls, and furniture. Instead of full ray tracing, illuminance adds a lightweight one-bounce diffuse term

$$I^{\mathrm{refl}}(\mathbf{p},t) = \sum_{s\in\mathcal{S}} \rho_s \bar{I}_s A_s^{\mathrm{rel}} e^{-\|\mathbf{p}-\mathbf{c}_s\|/\ell_s} \gamma_s(\mathbf{p}) V_s(\mathbf{p}), \qquad (3)$$

where surfaces $\mathcal{S}$ include room boundaries and active furniture faces, $\rho_s$ is reflectance, $\bar{I}_s$ is direct illuminance at surface center $\mathbf{c}_s$, $A_s^{\mathrm{rel}}$ is normalized area, $\ell_s$ is attenuation length, $\gamma_s$ is diffuse alignment, and $V_s$ is an obstruction term.

B. Sparse Calibration and Learning

Given predicted sensor value $\hat{y}_{i,m}$ and observed value $y_{i,m}$, residual $r_{i,m} = y_{i,m} - \hat{y}_{i,m}$ is used in two steps. First, active-device power scales are fitted by least squares over sensor residuals. Second, a trilinear correction is fitted:

$$\begin{aligned} C_m(x,y,z) = {} & c_{0,m} + c_{1,m}X + c_{2,m}Y + c_{3,m}Z + c_{4,m}XY \\ & + c_{5,m}XZ + c_{6,m}YZ + c_{7,m}XYZ, \end{aligned} \qquad (4)$$

where $X, Y, Z$ are normalized room coordinates. This cannot recover high-frequency turbulence, but it corrects bias, first-order gradients, and corner interactions from the eight observations. The trilinear form is deliberately matched to the sensing layout. With floor and ceiling sensors at all four horizontal corners, the eight observations constrain the eight coefficients without introducing a high-order surface that would overfit sparse points. The correction should therefore be interpreted as a low-frequency bias and gradient adjustment, not as a substitute for dense flow or lighting measurements. This design keeps the estimator identifiable from the available observations while still letting appliance geometry and dynamics remain in the explicit physical layer.

For non-networked devices, before/after sensor deltas supervise impact learning. The learning event is stored as an appliance-state record rather than as a free-text action label: it includes a record id, device name, applied device state, merged runtime device specifications, indoor baseline, outdoor boundary conditions, furniture state, elapsed time, and before observations. For an air conditioner, the applied state may include activation, mode, setpoint, fan speed or strength, outlet angles, and sweep settings. After observations from the same sensors complete the record. Given metric deltas $\Delta\mathbf{y}_m$ and appliance envelopes in matrix $X$, coefficients are estimated by

$$\boldsymbol{\beta}_m = \arg\min_{\boldsymbol{\beta}} \|\Delta\mathbf{y}_m - X\boldsymbol{\beta}\|_2^2. \qquad (5)$$

For residual learning, point features include normalized coordinates, elapsed time, indoor/outdoor conditions, base estimates, device activations, calibrated powers, appliance envelopes, and air-conditioner operating state such as mode, setpoint, fan strength, outlet angles, and sweep settings. The hybrid residual layer predicts $R_m(\mathbf{p},t;\theta_m)$ and returns $F_m^{\mathrm{hybrid}} = F_m + R_m$. This is the current-state spatial form used in the reported experiments. Its time-aligned $h$-step generalization is

$$\widehat{F}_m^{\mathrm{hybrid}}(\mathbf{p},t+h \mid \mathcal{I}_t) = \widehat{F}_m^{\mathrm{phys}}(\mathbf{p},t+h \mid \mathcal{I}_t)\widehat{R}_m(\mathbf{p},t+h \mid \mathcal{I}_t). \qquad (6)$$

Here, $h$ is forecast lead rather than a physical coefficient. The physics term carries $t+h$ because it is propagated from information available at forecast origin $t$ to produce the baseline at the same target time as the residual forecast. The information set $\mathcal{I}_t$ may include forecast boundary conditions and planned controls but excludes observations or truth residuals from $t+h$. The present spatial-estimation experiments correspond to $h=0$ and do not constitute a new multi-step forecasting evaluation. Temperature and humidity residual traces can be Fourier low-pass filtered before training because thermal and moisture responses are typically smoothed by heat capacity, air mixing, moisture exchange, and dehumidification. Illuminance uses raw residual labels because lighting, daylight, shadows, occlusion, and reflection can create physically meaningful short-timescale changes that low-pass filtering would remove.

The data flow in Fig. 3 is intentionally shared between training and runtime use. During learning, raw corner-sensor records, device events, outdoor boundary conditions, and scenario metadata are first time-aligned and normalized, then assembled into the same scenario state used by the service layer. This state is evaluated by the reduced-order estimator to produce base fields and residual labels. The pipeline then branches into least-squares impact learning from before/after sensor deltas and optional hybrid residual training from point-level residual samples, producing learned coefficients, validation summaries, and a checkpoint. At runtime, a user, web interface, or MCP client supplies the current baseline, outdoor conditions, device and furniture states, elapsed or steady-state time, and a query point or target zone. The service rebuilds the scenario state, evaluates the nominal fields, applies sparse correction and the optional residual checkpoint, and returns point or zone estimates. Action ranking is defined only after a decision sample scope is specified, either a single query point or a cluster/zone sample, and after a complete three-factor temperature, humidity, and illuminance target/tolerance vector is supplied. Without those inputs, the service returns estimates only and does not emit recommendations. Given scope and target, each candidate device action is converted into a counterfactual scenario and passed through the same inference path; actions are sorted by predicted comfort-penalty reduction.

Candidate control actions are evaluated as model-based counterfactuals. Let $S = \{\mathbf{p}_k\}_{k=1}^{K}$ be the decision sample

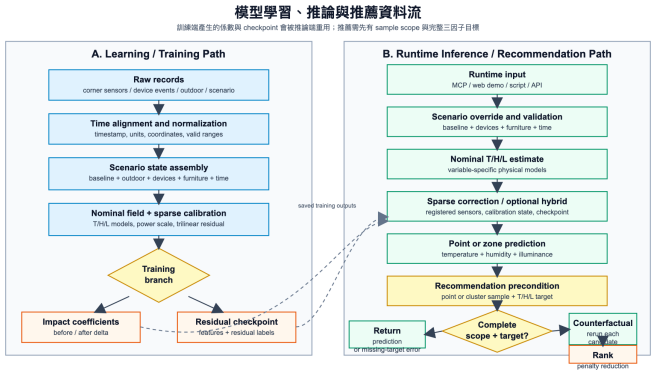



Fig. 3. Shared learning, inference, and recommendation data flow. Learned impact coefficients and the optional residual checkpoint are saved from the training path and reused by the runtime estimator.

scope, where $K = 1$ is a point query and $K > 1$ is a cluster or target-zone sample. For metric $m$, the scope value is $q_m(S) = K^{-1} \sum_k F_m(\mathbf{p}_k, t)$. A weighted comfort penalty is computed from normalized deviations beyond target tolerances for temperature, humidity, and illuminance, and each action is ranked by current penalty minus simulated post-action penalty. AC actions carry a fuller operating state rather than only an on/off flag: cooling, drying, heating, or fan mode; temperature setpoint; fan speed or numeric fan strength; horizontal and vertical outlet angles; and fixed or sweeping airflow. This is recommendation ranking, not closed-loop control; causal validation requires before/after intervention trials.

## IV. Implementation and Experimental Setup

The prototype is implemented as a zero-dependency Python package with modules for entities, scenarios, field simulation, impact learning, hybrid residual training, IDW comparison, action ranking, service APIs, an MCP endpoint, and a rotatable web demo. The shared service layer still supports the reproduction scripts and browser demo, while the MCP endpoint is intentionally narrowed to five runtime tools: environment initialization, elapsed-time or steady-state point sampling, before/after impact-learning records, direct window input, and point-target action ranking from registered devices. The ranking endpoint requires the point coordinates plus a complete temperature, humidity, and illuminance target; missing target fields produce no recommendation. The initialization tool registers the runtime state used by subsequent calls, including the base scenario, indoor baseline, outdoor boundary conditions, device and furniture states, AC mode/setpoint/fan/airflow controls when present, default elapsed or steady-state timing, and the estimator choice.

The service layer is intentionally kept separate from the estimator. A command-line script, the MCP endpoint, and the browser demo call the same scenario and sampling functions, so evaluation results are not produced by a separate code path from the interactive prototype. The browser interface supports metric switching, appliance toggling, AC mode/setpoint/fan-speed/outlet-angle controls, indoor baseline editing, season-weather-time presets, outdoor boundary overrides, point sampling, target-zone summaries, impact-learning output, and estimator switching between the base and hybrid residual fields when a saved checkpoint is available. This separation also reduces an implementation risk common in demonstration-driven digital twins: the visual interface can change without changing the numerical estimator, and the estimator can be tested from scripts without relying on browser state. The MCP layer is consequently treated as a deployment interface for the same service functions rather than as a separate modeling contribution. In the present paper, this matters because reported MAE values remain tied to reproduction scripts, while MCP point samples, learning records, direct window inputs, and action rankings expose the same runtime estimator without turning validation utilities such as IDW comparison or the full window matrix into MCP-facing claims.

All synthetic full-field experiments use the canonical room and $16 \times 12 \times 6$ grid, giving 1152 spatial samples per field. Scenarios are evaluated after an 18-minute settling interval with outdoor temperature 33.0°C, relative humidity 74.0%, and sunlight 32000 lux. The eight canonical scenarios are idle, AC only, window only, light only, AC+window, window+light, AC+light, and all active. Dense synthetic truth is generated by deterministic appliance power adjustments and sensor observations are synthesized from truth sensor predictions with fixed index-based perturbations: temperature adds $0.08((i \bmod 4) - 1.5)$, humidity adds $0.3((i \bmod 4) - 1.5)$, and illuminance adds $3.0((i \bmod 4) - 1.5)$. The main reproduction scripts are run_demo.py, run_hybrid_residual_experiment.py, run_submission_readiness_experiments.py, run_bedroom_weekly_simulation.py, and run_public_dataset_model_comparison.py.

The bounded indoor operating and application domain considered in this study is 20–30°C. Outdoor boundary inputs may lie outside that interval, as in the canonical hot-weather case, but they do not extend the claimed range of indoor target states or candidate deployment. Human-comfort ranking therefore uses target tolerances rather than assuming that smaller numerical error proves a need for equally narrow actuation.

Because no single dataset covers room geometry, dense 3-D ground truth, appliance state, and real sparse observations at once, the evaluation uses the layered claim boundary in Table I. We label the validation items as E1–E9 so that each numerical claim is tied to one evidence layer: E1–E6 are controlled synthetic or robustness experiments, E7 is a real-bedroom sparse-calibration check, E8 is a protocol for future before/after recommendation validation, and E9 is the public task-aligned benchmark. This separation is important because good performance

TABLE I
Validation Scope and Claim Boundary

| Evidence item | Supports | Boundary |
|---|---|---|
| E1–E4 synthetic full-field, IDW, ablation, and impact-learning checks | 3-D reconstruction, baseline comparison, and component contribution under controlled truth | Simulated room and deterministic perturbations |
| E5 48-case window matrix | Sensitivity to outdoor temperature, humidity, sunlight, and opening ratio | 34 in-domain cases; 14 out-of-domain stress cases |
| E6 hybrid residual splits | Residual learnability under default, no-Fourier, and leave-one-scenario-out settings | Scenario-family robustness, not arbitrary-room generalization |
| E7 seven-day bedroom snapshots | Sparse calibration improves a held-out pillow point | One reference point; no dense real 3-D ground truth |
| E8 recommendation intervention protocol | Defines measured improvement, success rate, and top-1 regret | Protocol only; not counted as completed validation |
| E9 SML2010 and CU-BEMS task-aligned benchmarks | External positioning on compatible time-series or zone-level prediction tasks | Not direct validation of the full spatial twin |

on one layer does not automatically validate the others.

## V. Evaluation

### A. E1–E4 Synthetic Reconstruction and Impact Checks

Across the eight canonical scenarios, the base estimator achieves average field MAE of 0.0474 for temperature, 0.1765 for humidity, and 2.0269 for illuminance. IDW gives 0.1723, 0.4633, and 54.9052, respectively. Table II summarizes the main comparisons and ablations. Removing appliance-aware feedback produces larger temperature and illuminance errors; removing one-bounce reflection or active-device calibration mainly hurts illuminance. The no-trilinear variant can be lower on dense synthetic humidity and illuminance because trilinear correction enforces sparse sensor consistency rather than guaranteed dense-field monotonic improvement. This last point is important for interpreting the ablation table. The trilinear correction is fitted from corner observations, so it is optimized for consistency with sparse sensors and for the real-bedroom calibration setting. It is not optimized directly against every dense grid point in the synthetic room. A lower dense-field MAE in the no-trilinear row therefore does not imply that sparse calibration is unnecessary; it indicates that sparse correction and dense synthetic truth are different objectives. For deployment, the estimator must respect measured corner observations first, then use the appliance-aware model to extrapolate the field between them.

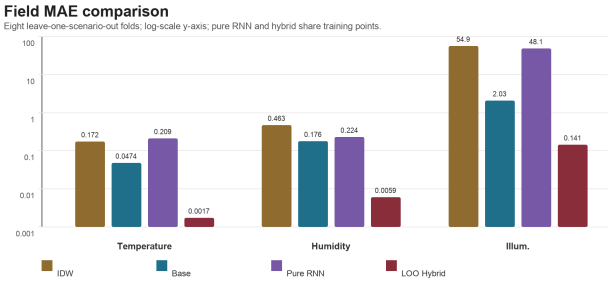



Fig. 4. Average field MAE for IDW, the base estimator, a pure RNN, and leave-one-scenario-out hybrid residual correction. The y-axis is logarithmic.

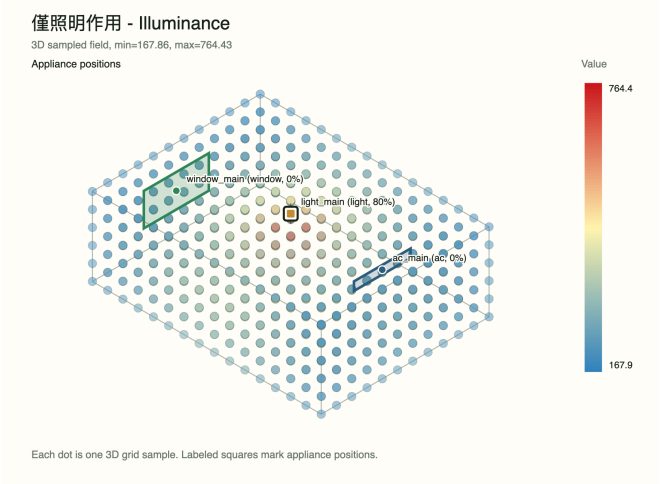



Fig. 5. Example 3-D illuminance field for the light-only scenario, showing localized lighting influence and reflected fill.

For impact learning, the AC-only scenario learns negative coefficients for temperature and humidity and near-zero illuminance effect. The light-only scenario learns a strongly positive illuminance coefficient, weak positive heat gain, and near-zero humidity effect. These directions match the intended physical interpretation of the appliances.

### B. E6 Pure RNN and Hybrid Residual Correction

The registered pure Elman RNN reads the same eight sparse observations as ordered spatial tokens and directly predicts absolute field values without physics, IDW, residual, or truth inputs. Each of eight folds trains on 672 points from seven scenarios and tests all 1,152 points of the held-out field. Its average MAE is 0.2091/0.2241/48.1422

TABLE II
Synthetic Field MAE and Robustness Summary

| Setting | Temp. | Hum. | Illum. |
|---|---|---|---|
| IDW baseline | 0.1723 | 0.4633 | 54.9052 |
| Pure RNN LOO | 0.2091 | 0.2241 | 48.1422 |
| Raw nominal | 0.1312 | 0.0842 | 3.5183 |
| No reflection | 0.0472 | 0.1762 | 2.4296 |
| No calibration | 0.0493 | 0.1772 | 3.3631 |
| No trilinear | 0.0446 | 0.0274 | 0.9849 |
| Full base | 0.0474 | 0.1765 | 2.0269 |
| LOO hybrid avg. | 0.0017 | 0.0059 | 0.1407 |

and it is lowest in 0/24 fold-factor comparisons; this adverse fixed-architecture result is retained. The default hybrid experiment yields 0.0020/0.0051/0.1370 on its 6/2 split, while the eight-fold LOO hybrid reaches 0.0017/0.0059/0.1407, or 96.41%/96.66%/93.06% reductions from the base estimator. These controlled results do not establish arbitrary-room generalization.

### C. E7 Real-Bedroom Snapshot Validation and E8 Intervention Protocol

We further evaluate calibration on a user-supplied bedroom dataset. The room is a 4.0 m by 4.6 m by 3.2 m bedroom with wall-mounted AC, southeast-facing window, ceiling light, desk lamp, bed, desk, and cabinets. The dataset covers 2026-04-14 to 2026-04-20, with 09:00, 15:00, 22:00, and 02:00 snapshots each day, for 28 cases. Each case provides eight corner observations, appliance activations, outdoor boundary information, and a pillow-location reference observation. Corner sensors are used for active-device calibration and trilinear correction; the pillow point is held out.

Table III shows that calibration reduces pillow-point MAE from 0.8967°C, 4.1286% relative humidity, and 309.0142 lux to 0.1676°C, 0.3939%, and 16.6450 lux. A 20,000-replicate paired date-block bootstrap (seed 20260726) yields mean reductions and 95% intervals of 0.7291 [0.4582, 1.0232]°C, 3.7346 [3.2005, 4.2524]%RH, and 292.3692 [288.3083, 297.0237] lux; 26/28, 28/28, and 28/28 snapshots improve. Seven overlapping leave-one-date-out folds retain positive minimum reductions of 0.6123°C, 3.5551%RH, and 290.5716 lux. This is an internal sensitivity check over one room, seven dates, and one held-out point, not independent replication or intervention success. Sleep snapshots use a 0 lux illuminance target with 5 lux tolerance so dark sleep conditions are not penalized. These snapshots validate sparse-sensor calibration and point-level estimation, not recommendation efficacy. A versioned E8 trial schema, empty field template, and deterministic analyzer now preregister the before/after endpoints. E8 currently has 0 completed real intervention trials; its summary remains NOT_EVALUATED, and all efficacy estimates remain null.

### D. E5 Window Matrix and E9 Public Benchmarks

The E5 window matrix has 48 time–weather–season cases and retains outdoor conditions as continuous inputs.

TABLE III
Real-Bedroom Snapshot Comparison

| Case | Temp. | Hum. | Illum. | Penalty |
|---|---|---|---|---|
| Raw pillow MAE | 0.8967 | 4.1286 | 309.0142 | – |
| Calib. pillow MAE | 0.1676 | 0.3939 | 16.6450 | 0.0911 |
| Morning calib. | 0.2372 | 0.3695 | 8.6247 | 0.2371 |
| Afternoon calib. | 0.1358 | 0.4890 | 11.2235 | 0.0812 |
| Night calib. | 0.0913 | 0.2193 | 46.6993 | 0.0459 |
| Sleep calib. | 0.2060 | 0.4979 | 0.0326 | 0.0000 |

TABLE IV
Selected Task-Aligned Benchmark Results

| Task | Metric | Pers. | LinReg | Proposed |
|---|---|---|---|---|
| S3 15min temp. | MAE | 0.233 | 0.093 | 0.071 |
| S3 15min temp. | Corr. | 0.000 | 0.909 | 0.950 |
| S3 60min temp. | MAE | 0.563 | 0.212 | 0.190 |
| S3 60min illum. | RMSE | 21.93 | 22.76 | 19.34 |
| S1 15min illum. | MAE | 3.42 | 4.02 | 5.35 |
| C2 15min illum. | MAE | 1.36 | 1.79 | 7.70 |
| S2 15min hum. | MAE | 0.20 | 0.21 | 0.83 |

Within the indoor 20–30°C domain, 34 target-zone results are in range and 14 remain only as stress cases.

SML2010 [14] and CU-BEMS [15] support only aligned time-series or zone tasks, not matched 3-D truth. Table IV retains favorable and adverse cases: persistence remains strong in high-inertia tasks, the registered RNN wins 0/12 cases, and the controlled Kalman/MA comparison splits 6/6 by variable. E11A screens 317 public BMC file-device cases inside 20–30°C; only 5 meet the minimum, persistence wins all 5, and thermal balance wins 0.

### E. Validity Considerations

Dense truth is synthetic; E7 covers one room, seven dates, and one held-out point, while public data validate aligned subtasks only. RNN and injected-noise Kalman results do not generalize to other architectures or deployed sensors. Claims remain within 20–30°C, and action rankings remain counterfactual pending interventions.

### F. Bounded AAU Enclosure Transfer

On 42 mapped AAU locations [16], [17], E11B (1,641 snapshots) rejects global IDW: nearest-neighbor MAE is 1.175 vs. 1.687 °C with 30/42 vs. 6/42 wins (H-ENC-02 unsupported). E11C (1,505 snapshots) lowers MAE from 1.301 to 1.223 °C but wins only 21/42 (H-ENC-03 unsupported). E11D role conditioning lowers global-mean MAE/RMSE from 2.3972/2.9748 to 1.6517/2.3648 °C, wins 30/42, and gives CI $[0.6867, 0.8124]$ °C; H-ENC-04 is predictive, not causal. E11E reaches MAE 1.0187 °C but P95 3.7699 °C and 25/42 wins; E11G reaches 0.8945/1.5415/3.1013 °C but 21/42, so both yield no_candidate_forwarded. Two-day E11H commissioning reaches MAE 0.4039 °C. Its frozen no-refit E11F map confirms MAE/RMSE/P95 0.3966/0.6723/1.2756 °C, 39/42 wins, and CI $[0.5851, 0.9274]$ °C. The decision is h_enc_05_supported_within_campaign: dates overlap development, while NTC hardware and cross-enclosure validity remain untested.

### G. BMC Data-Correction Chain

E12 stops before final-test access because six development files have fewer than 30 valid rows; E13 remains parser-invalidated. E14A/B preserve 4,038 source-correct rows and normalize three raw-unit files. On the already-opened E14C retrospective set, frozen load-aware ridge reduces MAE/RMSE/P95 from 4.0882/5.2087/12.0000 to 1.8054/2.8001/7.1146 °C, wins 13/14 runs, and yields a run-block MAE-gain interval of $[1.4271, 2.7939]$ °C. This supports candidate eligibility only. The separately preregistered E15 confirmation evaluates 14 previously unopened files (3,112 rows) once without refitting: MAE/RMSE/P95 decrease from 1.6939/1.9672/3.5000 to 0.9744/1.2056/2.3888 °C, with 12/14 run wins. Macro run MAE gain is 0.6134 °C with a 20,000-replicate run-block bootstrap 95% interval of $[0.2721, 0.9831]$ °C; all ten preregistered gates pass, supporting H-ENC-08. Runs 202308051757.csv and 202310252230.csv retain adverse MAE gains of $-0.5537$ and $-0.3011$ °C. Support is limited to same-server temporal/workload transfer; PC enclosures, NTC hardware, cross-server transfer, spatial fields, and control efficacy remain unverified. Enclosures are the active application direction within the broader single-room thesis scope.

### H. Exploratory Enclosure Control Simulation

The historical three-method SLSQP study is retained as v1 [18]; its opened holdout is not reused as unseen evidence. Version 2 compares fixed fan, equilibrium-feedforward PID, bounded gain-scheduled LQR [19], and OSQP MPC [20]. Shared assumed parameters, current observations, PWM/slew limits and safety overrides are fixed; each method receives three candidates on three calibration seeds, without equalizing CPU time. After freezing, 48 episodes (5,760 steps) cover validation, new synthetic holdout, three assumed plant variants and overload. Holdout macro tracking MAE is 2.9051/1.7713/1.7304/1.8071 °C, with fan-energy proxy 0.7200/0.9832/1.0034/1.0567 Wh, respectively. MAE, sampled overtemperature, command variation and numerical fallback are reported separately. Full-state access and known nominal parameters limit interpretation; designated plant variants do not establish cross-device generality. Thermal slack and successful numerical solves do not establish sufficient cooling or hardware safety. Physical identification, NTC measurements, throttling and whole-system energy remain unevaluated; E8 is unchanged.

### I. Sparse Thermal-Network Transfer Candidate

A separate temperature-only pilot follows the physics–sparse correction–action-ranking workflow. Conductive contact and convective airflow are represented separately, motivated by thermal design guidance [21], [22]. A hidden heat-source temperature is inferred from plate/air temperatures and fan observations. Three assumed rigs use identical bounded two-parameter calibration with known capacities and other coefficients. Across 108 episodes (19,440 steps), synthetic holdout source MAE for nominal/calibrated/corrected physics is 0.9610/0.6580/0.3352 °C. Fixed/PI/PID/one-step/six-step/six-step-without-correction tracking MAE is 1.5458/1.2219/1.2221/1.3969/1.2708/1.3644 °C. Six-step ranking improves over one-step ranking but does not outperform PI in tracking; sampled overtemperature averages 103.33 s per holdout episode. Finite candidate ranking borrows quadratic state/input costs and receding-horizon evaluation; it is not an LQR or full MPC solver. Kalman's foundational regulator study [23] motivates separating state estimation from control. Independent review corrected the raw-sensor warning interface and added scorer time/dynamics checks; input-only regression on the 72 opened closed-loop episodes changes no archived commands and is not new holdout evidence. Retrospective capacity analysis gives a 32.40 °C full-fan equilibrium for the contact-poor rig at 22 °C inlet and 3.35 W heat. For its six-step holdout, 150/360 held-input full-fan equilibria exceed 30 °C, versus 124 actual sampled overtemperature steps; steady capacity does not establish finite-horizon infeasibility or alter the original hypotheses. All parameters and fan energy are assumed; out-of-domain samples do not extend the validated room domain. Physical identification, hardware actuation and full three-factor transfer remain unevaluated.

### J. Integrated Sparse Enclosure Session

A separate offline temperature session integrates bounded four-parameter commissioning, sparse state estimation, LQR, linear MPC and action ranking. A fresh exploratory study uses 88 episodes (10,560 samples), four assumed rigs, 900 s commissioning per rig, capacity-ratio mismatch and one unmodeled direct source–air path. All controllers receive only plate/air/fan/inlet/power observations. Nominal/calibrated/corrected holdout source MAE is 1.0086/0.4681/0.3689 °C; correction improves the macro mean but only 2/4 pairs, failing the prespecified 3/4 coverage gate (H-ENC-51 not supported). Fixed/PI/PID/LQR/linear-MPC/one-step/six-step/no-correction tracking MAE is 1.7310/1.3471/1.3472/1.4872/1.3695/1.3764/1.3755/1.3943 °C. Six-step ranking fails the 5% improvement gate (H-ENC-52 not supported) and does not outperform PI. Its sampled overtemperature averages 3.75 s; all commands meet shared bounds/slew. Bounded LQR is not unconstrained optimal control; MPC uses a six-step affine linearization with soft thermal penalties. Independent dynamics and input-only replay verify every sample; all prior study files remain unchanged. This is commissioning-assisted assumed-model evidence, not hardware efficacy, arbitrary-family generality, measured energy or humidity/light/spatial transfer.

## VI. Discussion and Conclusion

The appliance-aware model outperforms IDW and the registered pure RNN under controlled truth, while residual calibration improves the sparse real-room point. In a fixed-budget, single-seed SML2010 check, GRU and LSTM are lowest-MAE in 0/12 cases; GRU beats vanilla RNN in 2/12 with median relative MAE reduction −12.88%, and LSTM in 0/12 with −11.37%, so the gated-recurrence hypothesis is unsupported. Public evidence remains task-bounded; dense measurement and interventions are future work, and physical PID control remains unevaluated. The separate four-controller comparison is exploratory assumed-model simulation; MPC does not outperform feedforward PID or LQR in its registered holdout.

## References


[1] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," IEEE Transactions on Industrial Informatics, vol. 15, no. 4, pp. 2405–2415, 2019.

[2] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," IEEE Access, vol. 8, pp. 108 952–108 971, 2020.

[3] A. S. Cespedes-Cubides and M. Jradi, "A review of building digital twins to improve energy efficiency in the building operational stage," Energy Informatics, vol. 7, p. 11, 2024.

[4] P. Bacher and H. Madsen, "Identifying suitable models for the heat dynamics of buildings," Energy and Buildings, vol. 43, no. 7, pp. 1511–1522, 2011.

[5] P. Hietaharju, M. Ruusunen, and K. Leiviskä, "A dynamic model for indoor temperature prediction in buildings," Energies, vol. 11, no. 6, p. 1477, 2018.

[6] E. J. Teshome and F. Haghighat, "Zonal models for indoor air flow - a critical review," International Journal of Ventilation, vol. 3, no. 2, pp. 119–129, 2004.

[7] B. Huljak, J. A. Acero, Z. H. Kyaw, and F. Chinesta, "Hybrid models for simulating indoor temperature distribution in air-conditioned spaces," Frontiers in Built Environment, vol. 11, p. 1690062, 2025.

[8] arealuser, "bmcdata: A dataset collected from server's baseboard management controller," GitHub dataset, commit 24904fa9a9bac49a3f6f3198bb04e1be5e2707ea, 2026. [Online]. Available: https://github.com/arealuser/bmcdata

[9] J.-H. Oh, S. Sfarra, and E.-J. Kim, "Hybrid modeling based on integrating simulation and operational data to improve indoor air temperature predictions, a controlled variable in digital twin models," Energy and Buildings, vol. 324, p. 114898, 2024.

[10] W. Qian, C. Li, H. Gao, L. Zhuang, Y. Lu, S. Hu, and J. Liu, "Estimating indoor air temperature and humidity distributions by data assimilation with finite observations: Validation using an actual residential room," Building and Environment, vol. 269, p. 112495, 2025.

[11] Y. L. Chen and J. Wen, "Application of zonal model on indoor air sensor network design," in Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2007, ser. Proceedings of SPIE, vol. 6529, 2007, p. 652911.

[12] D. Shepard, "A two-dimensional interpolation function for irregularly-spaced data," in Proceedings of the 1968 ACM National Conference, 1968, pp. 517–524.

[13] Model Context Protocol, "Model context protocol documentation," https://modelcontextprotocol.io/, accessed: 2026-04-10.

[14] P. Romeu-Guallart and F. Zamora-Martinez, "SML2010 [dataset]," UCI Machine Learning Repository, 2014.

[15] G. Chitalia and M. Pipattanasomporn, "CU-BEMS, smart building electricity consumption and indoor environmental sensor datasets," Scientific Data, vol. 7, p. 290, 2020.

[16] S. P. Melgaard, T. Juul, J. E. Nielsen, and R. L. Jensen, "Data from the AAU server room," 2026.

[17] O. M. Sarikaya et al., "Energy efficiency enhancement in two European data centers through CFD modeling," Scientific Reports, vol. 15, p. 24952, 2025.

[18] SciPy Developers, "Minimize method slsqp documentation," https://docs.scipy.org/doc/scipy/reference/optimize.minimize-slsqp.html, accessed: 2026-10-05.

[19] SciPy contributors, "solve_discrete_are: discrete-time algebraic riccati equation," https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.solve_discrete_are.html, accessed: 2026-10-05.

[20] OSQP Developers, "Model predictive control example," https://osqp.org/docs/examples/mpc.html, accessed: 2026-10-05.

[21] Texas Instruments, "Thermal design guide for dsp and arm application processors," https://www.ti.com/lit/an/sprabi3b/sprabi3b.pdf, 2017, sPRABI3B, revised August 2017; accessed 2026-10-06.

[22] Noctua, "Na-fd1 fan duct kit," https://www.noctua.at/en/products/na-fd1, accessed 2026-10-06.

[23] R. E. Kalman, "Contributions to the theory of optimal control," Boletin de la Sociedad Matematica Mexicana, vol. 5, pp. 102–119, 1960, second series. [Online]. Available: https://boletin.math.org.mx/pdf/2/5/BSMM(2).5.102-119.pdf